

[5353] - 107

**T.E. (Civil) (Semester - II) End Semester
PROJECT MANAGEMENT AND ENGINEERING
ECONOMICS
(2012 Pattern) (301008)**

Time :2.5 hours]

[Max. Marks :70

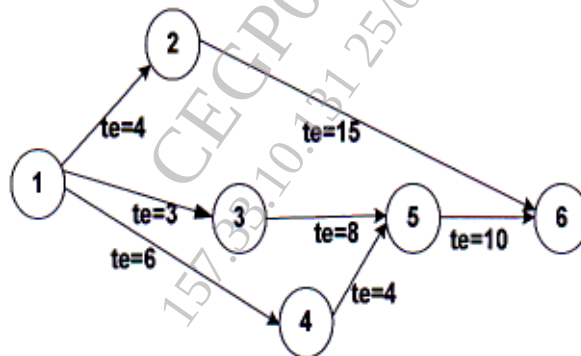
Instructions to the candidates:

Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10

- Q1)** a) Explain any five principles of scientific management. [5]
b) Define delegation of Authority. What are its advantages? [5]

OR

- Q2)** a) What are the domain areas of Project management? [5]
b) Find out expected project duration of the net work diag. given below. What are the chances of completing project in 21 weeks? Mark critical path. [5]



- Q3)** a) Draw the network diag., find Project duration and mark critical path By using following data [6]
i) Activity A & B are starting activities, requires 9 and 6 days to complete resp.
ii) Activity C follows activity B, activity C takes 2 days to complete.
iii) Activity D succeeds activity C and A, activity D take 5 days to complete
iv) Activity D is terminal activity
b) What are the methods of planning and scheduling? Explain any one. [4]

P.T.O.

OR

- Q4)** a) List out step by step procedure of network compression. [4]
b) Find out the cost of project before crashing? What will be the change in cost of project by crashing it to minimum time? Consider indirect cost is Rs.100/-day [6]

Activity	NT (days)	CT (days)	NC (Rs.)	CC (Rs.)
10-20	6	2	1000	1200
10-30	8	5	1500	1800
10-50	13	10	2000	2450
20-30	4	3	500	550
20-40	10	8	1600	2000
30-50	7	4	1200	1500

- Q5)** a) Explain the following [8]
i) Annuities and their types
ii) Law of diminishing marginal utility
b) Discuss importance of economics in construction industry [4]
c) What do you understand by Law of substitution? [4]

OR

- Q6)** a) Write a short note on: [8]
i) Demand curve and factors affecting on it
ii) Supply curve and factors affecting on it
b) Define Goods, Wants, assets, liability with the help of example. [4]
c) Ms.Madhuri invests Rs. 100000/- in a bank at nominal interest rate 12% for 7 years. The compounding is Monthly. Find future amount she will get after 7 yrs. [4]

- Q7)** a) What are the objectives of material management and functions of purchase dept.? [6]
 b) Define inventory and explain any one technique of controlling inventory. [6]
 c) Define IFR, ISR and injury index with formulas [6]

OR

- Q8)** a) What do you know about the indent? Explain process of material issue and return of site you visited. [6]
 b) What points should you consider while making safety programme of construction site? [4]
 c) "Sakalp groceryshoppee" carries the following items .Segregate the items based on their annual usage and plot ABC curve. [8]

Sr. No.	Item	Annual Consumption	Cost in Rs. per unit
1	Sugar	150 bags	4000/-
2	Wheat	400 quintal	2000/-
3	Moog dal	3000 kg	60/-
4	Tur dal	4500 kg	100/-
5	Oil sunflower	3000 lit.	90/-
6	Chocolates	3000 pkts.	50/-
7	Soap	10,000 nos.	10/-

- Q9)** a) What are the different types of appraisals required to undertake any Project? Explain any one in detail. [6]
 b) Explain break even analysis with help of figure showing breakeven point, break even sales, angle of incidence and margin of safety. [6]
 c) Write a short note on any one [4]
 i) Payback period
 ii) Role of PMC

OR

- Q10)** Write a short note on any four [16]
 a) IRR method
 b) BC ratio method
 c) Detailed project report (DPR)
 d) NPV method
 e) Pre tendering and post tendering

